



BIOINFORMATICS

Bioinformatics Bootcamp 2026

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Assignment 01

Topic: Biological Databases and Bioinformatics Applications

Question 1

Question: A researcher has discovered a DNA sequence that may be associated with a rare genetic disorder. Before conducting laboratory experiments, they want to determine whether this sequence has been reported previously.

Answer: I would recommend using GenBank, NCBI BLAST, and OMIM. GenBank contains millions of nucleotide sequences submitted by scientists worldwide. By searching the sequence in GenBank, the researcher can determine whether a similar sequence has already been reported. BLAST can compare the sequence against existing records and identify close matches. OMIM provides information about genes linked with inherited diseases and genetic disorders. For example, if the sequence shows similarity to a gene known to cause cystic fibrosis, OMIM can provide detailed disease information. Using these databases before laboratory work saves time, reduces cost, and helps researchers understand the biological significance of the sequence.

Question 2

Question: A scientist is studying a disease-causing bacterium and wants to compare its genome with related bacterial species to identify conserved genes.

Answer: For this analysis, I would use NCBI Genome, RefSeq, and Ensembl Bacteria. These databases contain complete bacterial genome sequences and annotations. Scientists can compare genomes from different species to identify conserved genes that perform essential biological functions. Conserved genes are often involved in critical cellular processes and remain similar throughout evolution. For example, genes involved in DNA replication are commonly conserved among many bacterial species. Comparative analysis can also reveal unique genes responsible for pathogenicity or antibiotic resistance. These databases provide reliable genomic information and are highly useful for comparative genomics studies.

Question 3

Question: A biotechnology company has isolated a protein from a newly identified microorganism and wants to investigate whether similar proteins exist in other species.

Answer: I would recommend UniProt, NCBI Protein, and the Protein Data Bank (PDB). UniProt provides detailed information about protein sequences, biological functions, and annotations. NCBI Protein contains a large collection of protein records from different organisms. PDB provides three-dimensional structures of proteins. For example, if the isolated protein resembles an enzyme found in bacteria and fungi, these databases can reveal sequence similarities and structural relationships. This information helps researchers predict protein function, understand evolutionary relationships, and evaluate possible industrial or medical applications.

Question 4

Question: A scientist has identified an unknown protein sequence and wants to determine its function.

Answer: To determine the function of an unknown protein sequence, I would use UniProt, InterPro, Pfam, and PDB. UniProt provides protein descriptions and functional information. InterPro and Pfam identify conserved domains and protein families that often indicate biological function. PDB helps compare protein structures with known proteins. For example, if the sequence contains a kinase domain identified through Pfam, the protein may be involved in cell signaling. By combining sequence, domain, and structural information, researchers can make accurate predictions about protein function before experimental validation.

Question 5

Question: You have been provided with an unknown biological sequence but are not told whether it is DNA, RNA, or protein. Describe the step-by-step approach you would follow to identify the sequence using biological databases.

Answer: First, I would examine the sequence composition. If it contains amino acid symbols, it is likely a protein sequence. If it contains only nucleotide bases, it may be DNA or RNA. Next, I would use BLASTN for nucleotide sequences or BLASTP for protein sequences. After obtaining matches, I would search databases such as GenBank, UniProt, and RefSeq for detailed annotations. I would then analyze the sequence using Pfam and InterPro to identify domains and functional regions. For example, if the sequence matches a known hemoglobin protein, UniProt can provide information about its biological role. This systematic approach helps accurately classify and understand unknown biological sequences.